Curriculum Vitae
Xuanang LEI

leixuanang@outlook.com · github.com/Leon-LXA

Education

ETH Zurich, Zurich, Switzerland Sept. 2022- Sept. 2025

Master in Robotics, Systems and Control. GPA: 5.76/6.00, rank:30/180
Selected major courses:

- *Introduction to Machine Learning(6/6)*, *Computer Vision(5.75/6)*, *Planning and Decision Making for Autonomous Robots(5.75/6)*, *Vision Algorithms for Mobile Robotics(5/6)*, *Probabilistic Artificial Intelligence(5.5/6)*
- *Foundations of Reinforcement Learning(5.5/6)*, *Model Predictive Control(5.75/6)*
- *Robotics Summer School*

Tsinghua University, Shenzhen, China Jun. 2021- Sept. 2022

Research Assistant, Advisor: Prof Xueqian Wang

Harbin Institute of Technology (HIT), Shenzhen, China Sept. 2018- Jun. 2022

B.E. in Automation. GPA: 92.9/100, rank: 1/125
Academic awards and scholarships:

- National Scholarship(2020)
- Scholarship of university * 3(2018-2021)
- First prize in ABU ASIA-PACIFIC ROBOT CONTEST 2020
- Excellent Student Leader(2018-2019)

Publication

- **Dynamic Control Barrier Function-based Model Predictive Control to Safety-Critical Obstacle-Avoidance of Mobile Robot**
  Zhuozhu Jian*, Zihong Yan*, **Xuanang Lei**, Zihong Lu, Bin Lan, Xueqian Wang, Bin Liang
  IEEE International Conference on Robotics and Automation (ICRA), 2023.
- **A Synthetic-Driven Vision System for Assembly Step Recognition**
  Hui Zhang*, **Xuanang Lei***, Rui Wang, Julian Ferchow, Mirko Meboldt
  IEEE 22nd International Conference on Automation Science and Engineering (CASE), 2026.
- **Quadrotor External-Wrench Estimation via Physically Structured Residual Learning**
  Xuanang Lei, Yunfan Ren
  To be submitted to *IEEE Robotics and Automation Letters (RA-L)*.

Work Experience

Inspire AG, Zurich, Switzerland Feb. 2025 - Sept. 2025

R&D Engineer Intern Supervisor: Dr. Julian Ferchow

- Developed an assembly motion recognition and real-time error detection system with **Bossard AG** and **Intelliact AG**.
- Validated the system through field tests at **Stäubli** and **Mettler-Toledo** factories.

Viterra Robotics, Chengdu, China Apr. 2026 - Oct. 2026

Robotics Algorithm Engineer

- Established an end-to-end sim-to-real development workflow for quadrotor autonomy, spanning simulation, onboard deployment, and real-world flight validation.

- Developed autonomous exploration and motion planning algorithms for quadrotors navigating complex, GPS-denied environments.
- Conducted flight-dynamics identification and control-oriented modeling, and developed physics-informed learning methods for external force and torque estimation.

RESEARCH EXPERIENCE

Assembly Motion Generation for Action Recognition | ETH Zurich Aug. 2024- Feb. 2025

Master Thesis Student (Grade: 6.0/6.0) Advisor: Prof. Dr. Mirko Meboldt at Pd|z ETH.

- RL-based training of scalable dexterous grasp generation with GraspXL framework
- Synthesize assembly motion with grasping hand in simulation.
- Action recognition training with synthesized assembly motion and label
- Monitor user interface to detect possible assembly mistakes in Mettler-Toledo and Staubli.

Adversarial Learning via Differentiable Simulation | ETH Zurich Nov. 2023- Mar. 2024

Semester Project Student Advisor: Prof. Fan Shi at ETH AI Center.

- Adversarial attack algorithm to disrupt the policy of quadruped robot when the gradient information is accurate in differentiable simulation
- Projected Gradient Descent on joint-state information to make legged robot fall over.

Adaptive Mechanism Design in Sequential Social Dilemmas | ETH Zurich Apr. 2023- Jun. 2023

Student Advisor: Dr. Vinzenz Thoma at D-INFK, ETH AI Center.

- Re-implement the AMD algorithm using ray.rllib. Compatible with any multi-agent environment. Validated by using the prisoner's dilemma and further modify the algorithm to accommodate complex environments
- Provide open-source implementations of the Wolfpack and Gathering environments.

Navigation of Mobile Robots on Rough Terrain | Tsinghua University Jun. 2021- Sept. 2022

Research Assistant Advisor: Prof. Wang Xueqian at the School of Control Science and Engineering, Tsinghua University.

- Real-time SLAM and navigation on SCOUT|AgileX Robotics on extremely rough terrain
- Local planning(Collision Avoidance) with MPC, CBF and global planning by PF-RRT*
- Mapping with Point Cloud of the complex terrain

LEADERSHIP IN COMPETITION

ABU ROBOCON Team: WTR Dec. 2019- Mar. 2021

Captain Advisor: Yunjiang Lou, Professor and Dean at Mechanical Engineering and Automation, HIT.

- Led the team and won the first prize in the ABU ROBOCON 2020, as the top 16 of 75 participating universities.
- Joined the team and won the national second prize in the ABU ROBOCON 2019 in the first year
- Established a robotics forum on campus to promote technological advancement, which attracted over 300 students to participate

SKILLS

- **Programming & Software:** 8 years of experience in C++, Python, MATLAB; expert in Robot Operating System (ROS) and OpenCV; abundant embedded programming experience (STM32, Arduino).
- **Robotics:** Motion planning, SLAM, real-time navigation, grasping algorithms.
- **Machine Learning & AI:** Reinforcement learning, probabilistic AI, computer vision, action recognition, adversarial learning.
- **Simulation & Rendering:** Proficient in Blender for photorealistic rendering, Issac Gym, RaiSim, Webots, CoppeliaSim, and SolidWorks for 3D modeling and simulation.